

AgriConnect — SIH26132 Solution Document

Strengthening Market Linkages and Price Discovery for Farmers

Organization: Government of Maharashtra | **PS Number:** SIH26132 | **Theme:** Agriculture, FoodTech & Rural Development | **Category:** Software

1. Problem Statement (Official)

Title: Strengthening market linkages and price discovery for farmers

Problem Description (as issued by Government of Maharashtra):

Many farmers, especially smallholders and producer groups, have limited visibility of current and expected prices across nearby markets, processors, institutional buyers and digital trading channels. Information on quality specifications, demand, logistics, storage, payment reliability and buyer credentials may be fragmented. Farmers may sell immediately after harvest because of liquidity or storage constraints and may have weak bargaining power. Buyers, meanwhile, may struggle to aggregate consistent volumes and verify quality. The challenge is to improve transparent price discovery and create reliable, efficient linkages from farm gate to suitable buyers.

Expected Solution / Outcome (as issued):

A market-intelligence and transaction enablement solution that aggregates mandi prices, buyer demand, quality requirements, arrival volumes, transport and storage options; provides localised price trends and sale-window recommendations; matches farmers/FPOs with verified buyers; enables lot creation, quality grading, digital offers, logistics coordination and payment tracking; and supports dispute or grievance processes. Expected outcomes include improved farmer price realisation, reduced information asymmetry, lower transaction cost, stronger FPO aggregation, reduced post-harvest loss, more reliable buyer sourcing and transparent transaction records.

2. Problem in Simple Terms

Think of it from a farmer's point of view:

You've just harvested your crop. You don't really know what price it's fetching at the mandi 20km away versus the one you always sell at — the information exists somewhere, but not in a form you can quickly check.

Even if you did know the price, you don't know if it's about to go up (worth waiting) or down (better to sell now).

You have no cold storage, so you often *have* to sell fast, even at a bad price — you have no cushion to negotiate.

You don't know which buyers pay reliably and which don't.

Even if a good buyer exists, you alone might not have enough produce to meet their minimum order — but you don't have an easy way to team up with nearby farmers who grow the same thing. If a deal goes wrong (bad payment, quality dispute), you have no formal way to raise it.

Now think of it from a buyer's side (a trader, processor, or institutional buyer):

You need a **consistent, verified-quality volume** of a crop — but sourcing that means dealing with dozens of scattered small farmers individually, with no guarantee of quality consistency.

You have no easy way to place a demand ("I need 5 tons of grade-A onions by Friday") and have it reach the right farmers.

In one sentence: both sides of the transaction are operating with incomplete information and no efficient way to find and trust each other — and that gap is where farmers lose money and buyers lose reliability.

3. Why This Is Still an Open Problem (Market Gap)

It's important to know this *isn't* a totally unsolved space — government platforms already exist:

Agmarknet / eNAM already provide live mandi price data and even online bidding.

Private apps (DeHaat, Kisan Suvidha, Khetihaar, Farmer.in) already show prices and, in some cases, marketplace features.

So what's actually missing?

Raw prices ≠ a decision. No existing tool tells a farmer "*sell here, sell now*" — factoring in transport cost, quality, and price trend, in one clear answer.

Access is still app/web-first. Farmers with low literacy or poor connectivity are underserved — this is explicitly acknowledged as a persistent gap in eNAM adoption.

No lot-level transaction workflow. Existing tools stop at "here's a price" or "here's a listing" — none carry a farmer through lot creation → grading → offer → logistics → payment → dispute resolution, which is exactly what this PS is asking for.

No dynamic aggregation for non-FPO farmers. Formal FPOs solve bargaining power for their members; there's no lightweight way for *any* group of nearby smallholders to pool an order on demand.

No transaction transparency layer. Buyer reliability, payment tracking, and grievance handling are not addressed by any current price-display tool.

Our positioning: Don't rebuild price data (that already exists and works) — build the **decision + transaction layer on top of it** that nothing else currently offers.

4. Proposed Solution — AgriConnect

AgriConnect is a market-intelligence and transaction-enablement platform with two entry points:

Farmer/FPO side: WhatsApp or voice-call based (low-literacy, low-connectivity friendly)

Buyer/Admin side: Web dashboard (for buyers, FPOs, and government reviewers/judges)

4.1 End-to-End Solution Flow

Stage	What it does
1. Farmer/FPO input	Voice call or WhatsApp message — no app download required
2. NLP intake & lot creation	Extracts crop, quantity, quality, location; creates a structured "lot" record
3. Price advisory engine	Pulls Agmarknet/eNAM data, computes net price per nearby mandi, shows trend
4. Quality grading	Self-declared or photo-assisted grading, mapped to Agmarknet's standard grades
5. Sell-now / hold recommendation	Plain-language recommendation based on stages 3–4
6. Buyer demand & matching	Buyers post demand with specs; system matches lots to demand
7. Ad-hoc + FPO aggregation	Pools nearby smallholders to meet buyer minimum order quantities
8. Digital offer & negotiation	Buyer sends offer on a lot; farmer accepts or counters
9. Logistics & storage coordination	Surfaces nearby transport/storage options once a deal is agreed
10. Payment tracking	Status tracker: offer accepted → pickup confirmed → payment initiated → received
11. Dispute/grievance handling	Farmer or buyer can flag an issue tied to a transaction ID

4.2 What Makes This Different (Innovation Summary)

Decision layer, not just a data layer — net-price calculation + sell/hold advisory, which no current platform offers together.

Voice-first, not voice-added — designed around a phone call/voice note as the primary interface, not a translated afterthought.

On-demand aggregation without FPO membership — dynamic clustering of nearby smallholders for any transaction, not just formal FPO members.

Full transaction lifecycle — lot creation through payment tracking and disputes, directly matching what the PS explicitly asks for (most competitors stop at "show a price" or "list an item").

Built on government data, not competing with it — uses Agmarknet/eNAM as the price source, making it realistically deployable with state backing.

5. Roadmap (Phased)

Phase 0 — Idea Submission (now → 30 September 2026)

Goal: submit a strong, clear idea on the SIH portal.

Finalize problem understanding and solution scope (this document)

Prepare idea submission write-up: problem, solution, tech stack, feasibility, impact

Create a basic pitch deck / explainer video if required by SIH round-1 format

Identify and lock final team roles

Phase 1 — Pre-Hackathon Build-Up (if shortlisted, before the SIH finals hackathon)

Goal: have a working MVP skeleton ready before the 36-hour build, since real hackathon time is short.

Set up repo, environments, and API access (Agmarknet/eNAM keys, WhatsApp Business API/Twilio sandbox)

Build and test the Price Advisory Engine independently (this is your core differentiator — get it right first)

Build a basic data model for lots, buyers, offers, transactions

Build a rough NLP intake pipeline (even keyword/slot-based extraction is fine to start)

Phase 2 — Hackathon Build (during SIH finals, ~36 hours)

Goal: working end-to-end demo covering stages 1–8 at minimum (see implementation plan below).

Phase 3 — Stretch / Post-Hackathon Vision (for pitch, not necessarily built)

Full logistics/storage marketplace integration

Real payment gateway integration

ML-based price forecasting (beyond simple trend/moving average)

State-wide rollout plan, integration proposal with Maharashtra's agriculture department

6. Step-by-Step Implementation Plan (Team-Facing)

Suggested Team Roles (for a typical 6-member SIH team)

Role	Responsibility
Team Lead / PM	Coordinates timeline, owns the pitch narrative, talks to mentors/judges
Backend Dev 1	Price Advisory Engine — Agmarknet/eNAM integration, net-price + trend logic
Backend Dev 2	Lot/offer/transaction data model, matching & aggregation logic
Frontend Dev	Web dashboard (buyer/admin side)
NLP/Conversational Dev	WhatsApp/voice intake flow, language extraction

Role	Responsibility
Design/Presentation	UI/UX for both farmer and buyer flows, final pitch deck, demo video
<i>(Adjust based on your actual team size and skills — roles can double up.)</i>	
Step-by-Step Build Order	
Step 1 — Data foundation (Day 1, morning)	
Get Agmarknet/eNAM API access working; pull sample price data for 3–5 crops relevant to Maharashtra (onion, cotton, soybean, tur dal, tomato)	
Design the core data schema: Lot , Farmer , Buyer , Offer , Transaction	
Stand up backend project structure and database	
Step 2 — Price Advisory Engine (Day 1, midday)	
Implement net-price calculation: mandi price – transport cost estimate	
Implement simple trend detection (7–14 day moving average) for sell-now/hold signal	
Test with real data for your chosen sample crops — this is your headline feature, get it solid before anything else	
Step 3 — Farmer intake flow (Day 1, afternoon)	
Build WhatsApp bot (or simplified voice-note upload + transcription) that extracts: crop, quantity, quality, location	
Connect intake output to Price Advisory Engine — farmer should get a price recommendation back in the same conversation	
Step 4 — Lot creation & quality grading (Day 1, evening)	
Turn a completed intake into a structured "lot" object	
Add a simple grading step — checklist-based or photo-upload placeholder mapped to standard grade categories	
Step 5 — Buyer dashboard: demand posting & matching (Day 2, morning)	
Build web dashboard where a buyer can post demand (crop, quantity, quality spec, price offered)	
Implement basic matching: system surfaces lots that fit a buyer's posted demand	
Implement basic aggregation: if no single lot meets the buyer's minimum quantity, cluster nearby smaller lots	
Step 6 — Offer & negotiation flow (Day 2, midday)	
Buyer sends an offer on a matched lot (or aggregated cluster)	
Farmer (via WhatsApp) receives the offer and can accept or reject	
Store offer status in the transaction record	
Step 7 — Logistics/storage + payment tracking (Day 2, afternoon — keep simple)	
Logistics/storage: a static directory lookup (nearest known storage/transport option) is enough — don't over-build	

Payment tracking: a simple status field (offer accepted → pickup confirmed → payment initiated → payment received) shown on both dashboard and WhatsApp — no real payment gateway needed for the demo

Step 8 — Dispute flag (Day 2, late afternoon)

Add a simple "raise an issue" action tied to a transaction ID, visible on the buyer dashboard as a flagged-item list

This can be UI-only for the demo — you just need to show the concept works

Step 9 — Polish, test, and rehearse (Day 2, evening)

Run through the full flow end-to-end at least 3 times as a team

Fix obvious bugs, clean up the UI

Prepare the pitch: lead with the *gap* (why this isn't already solved), then the *demo* (live, not slides), then the *impact* (price realisation, reduced post-harvest loss, FPO strengthening)

What to Fully Build vs. What to Mock for the Demo

Fully build (core, must work live)	Mock or simplify (show concept, don't over-engineer)
Price Advisory Engine (real Agmarknet/eNAM data)	Logistics/storage (static list is fine)
WhatsApp/voice intake	Payment gateway (status tracker only, no real money movement)
Lot creation & matching	Full quality-grading ML (checklist/photo placeholder is fine)
Buyer dashboard + offer flow	Dispute resolution backend (UI flag is enough)

7. Suggested Tech Stack

Frontend (buyer/admin dashboard): React / Next.js

Farmer-side interface: WhatsApp Business API (or Twilio sandbox for demo) / basic IVR simulation

Backend: Python (FastAPI) or Node.js — FastAPI recommended if trend/price logic gets data-science-heavy

NLP intake: LLM-based structured extraction (prompt-based slot filling) or a lightweight intent/entity model if offline-capable NLP is preferred

Database: PostgreSQL (relational data — lots, offers, transactions — benefits from structured queries and joins)

Price data source: Agmarknet API / eNAM public data

Hosting for demo: Vercel (frontend) + Render/Railway (backend) — fast to deploy, judge-accessible via a live link

8. Expected Impact (for pitch narrative)

Directly mirrors the PS's stated expected outcomes:

Improved farmer price realisation — via net-price comparison and sell/hold timing

Reduced information asymmetry — single source combining prices, demand, quality specs, logistics

Lower transaction cost — fewer intermediaries via direct buyer matching

Stronger FPO aggregation — formal FPO support plus ad-hoc pooling for non-members

Reduced post-harvest loss — better sale-timing guidance reduces forced distress sales

More reliable buyer sourcing — verified buyer profiles and transaction history

Transparent transaction records — offer, payment, and dispute status all tracked and visible

9. Risks & Mitigations

Risk	Mitigation
Agmarknet/eNAM API access delays or rate limits	Cache sample data locally early; have a fallback static dataset for demo day
WhatsApp Business API approval takes time	Use Twilio sandbox or a simulated chat UI as backup for the demo
Scope is too large for 36 hours	Strictly follow the "fully build vs. mock" table above — don't let scope creep into logistics/payment
NLP intake struggles with regional language/accents	Start with a smaller set of well-tested sample phrases for the live demo; keep a text-input fallback
Judges ask "how is this different from eNAM"	Have the gap-analysis section (Section 3) ready as a crisp one-line answer: <i>"eNAM tells you the price — we tell you what to do and connect you to a verified buyer, end to end."</i>

Document prepared for team alignment on SIH26132 — AgriConnect. Update this as the solution evolves through Phase 0-2.